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Abstract

This paper investigates the use of Angle of Arrival (AoA) as an authentication measure in analog array multiple-input multiple-output (MIMO) systems. We examine how a base station (BS) equipped with an analog array can authenticate users based on AoA estimated from certified pilot transmissions, while analyzing vulnerabilities to active attackers who manipulate their transmitted signals for impersonation. Building on this baseline framework using a Uniform Linear Array (ULA), we explore two independent extensions to enhance authentication robustness. First, we implement a 3D AoA estimation approach using a Uniform Planar Array (UPA) at the BS, enabling angle estimation along both azimuth and elevation axes, thereby increasing spatial resolution for improved impersonation detection. Second, we extend the system to a cooperative multi-BS defense strategy, where cross-verification among geographically distributed BSs further strengthens resistance against sophisticated attacks. Across these configurations, we analyze various adversarial scenarios with increasing levels of side information and evaluate system resilience using one-class classifiers. Our results show that both the 3D and multi-BS extensions significantly improve the reliability of authentication, especially under sophisticated spoofing attempts targeting spatial domain features. Index Terms—Physical layer authentication, AoA-based authentication, 3D AoA estimation, UPA, multi-base station systems, impersonation attack, spatial diversity.

Chapter 1

Introduction

1.1 Motivation and Background

Motivation and Background In the evolving landscape of wireless communications, ensuring secure user authentication at the physical layer has become increasingly vital. Traditional cryptographic methods, while effective, often suffer from computational overhead and vulnerability to key-based attacks. As a complementary or alternative solution, physical layer authentication schemes have garnered significant interest, particularly those leveraging spatial characteristics inherent to wireless channels. PLA operates directly at the physical layer, using properties such as received signal strength, channel impulse response, or spatial direction (e.g., Angle of Arrival) to verify the identity of a transmitter without relying on shared cryptographic keys. Among such approaches, Angle of Arrival (AoA)-based authentication has emerged as a promising technique for differentiating between legitimate and malicious users by utilizing the spatial signature of incoming signals.

1.2 Baseline Framework: Single-BS with ULA

This work investigates the feasibility and robustness of AoA as a physical layer authentication mechanism in analog array multiple-input multiple-output (MIMO) systems. We begin with a baseline model in which a single base station (BS) equipped with a Uniform Linear Array (ULA) estimates the AoA of pilot signals transmitted by users. Authentication is performed by comparing the estimated spatial signatures with expected values to detect potential impersonation attacks. This foundational framework provides a practical means for spatial verification but is inherently limited by the one-dimensional angular resolution of the ULA.

1.3 3D AoA Extension: Single-BS with UPA

To enhance authentication fidelity, we introduce a three-dimensional extension of the baseline system by employing a Uniform Planar Array (UPA) at the BS. This configuration allows for simultaneous estimation of both azimuth and elevation angles, providing finer spatial resolution and increased robustness against adversaries attempting to mimic the spatial signature of legitimate users. This 3D AoA approach effectively adds an additional axis of verification, thereby expanding the authentication capabilities of the system.

1.4 Multi-BS Extension: Cooperative Authentication

Subsequently, we extend the framework further by incorporating a multi-base station (multi-BS) architecture. In this setup, spatial diversity is exploited by enabling multiple geographically separated BSs to collaboratively authenticate users. Through cross-verification of AoA estimates across the BSs, the system gains enhanced resilience to sophisticated impersonation attacks, including those where the adversary possesses substantial side information.

1.5 Adversarial Analysis and Evaluation

Across all configurations—single-BS ULA, single-BS UPA, and coordinated multi-BS—we systematically analyze adversarial scenarios with escalating levels of complexity. Employing one-class classifiers as our detection methodology, we rigorously assess the system’s ability to identify unauthorized transmissions. Our findings indicate that the 3D extension and multi-BS coordination each independently contribute to significant improvements in authentication reliability, with their combined effect offering a robust defense against advanced spoofing strategies.

Chapter 2

Existing Work

2.1 Overview of Physical Layer Authentication (PLA)

Physical Layer Authentication (PLA) has gained traction as a lightweight, context-aware security mechanism that exploits the inherent characteristics of wireless communication channels and devices. Unlike cryptographic methods, PLA does not rely on computationally expensive operations or pre-shared secrets. Instead, it leverages physical attributes—such as channel responses, signal angles, and hardware imperfections—to establish user authenticity.

2.2 Signal-Level and Channel-Based Approaches

Early research in PLA primarily centered on signal-level characteristics. Techniques based on channel impulse responses (CIR) and carrier frequency offsets (CFO) were among the first to be explored for differentiating users. These features were found to be device- and location-specific, offering a basic level of identity assurance. Radio frequency fingerprints, induced by hardware imperfections, also gained attention as unique device identifiers.

2.3 Channel State Information (CSI)-Driven Techniques

With advancements in MIMO technology, Channel State Information (CSI) became a key enabler for PLA. CSI provides fine-grained information about the multipath characteristics of the wireless channel, which can be uniquely associated with a device's position and transmission properties. These attributes allow for high-resolution authentication and have been widely used in practical implementations.

2.4 AoA-Based PLA and Spatial Awareness

Angle of Arrival (AoA) estimation added a new spatial dimension to PLA by allowing systems to infer the direction of incoming signals. This spatial awareness made it harder for attackers to spoof legitimate transmissions without physically replicating the user’s location. However, it also introduced challenges—particularly from sophisticated adversaries capable of directional spoofing or exploiting beamforming to mimic legitimate angles.

2.5 Machine Learning for PLA

To address spoofing and generalization challenges, researchers have integrated machine learning models, particularly Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), and hybrid models. These methods analyze patterns in physical-layer features to detect anomalies and classify transmissions as genuine or malicious. One-class classifiers, in particular, have shown promise in scenarios where only legitimate user data is available for training.

2.6 Protocol-Level Innovations

Dynamic challenge-response mechanisms and zero-round-trip-time (0-RTT) multifactor authentication schemes represent another major advancement. These strategies reduce protocol latency while maintaining strong security guarantees. They often combine multiple PLA features—e.g., combining CSI and AoA—to increase spoof resilience.

2.7 Applications and Emerging Domains

PLA techniques are not limited to conventional cellular networks. Specialized applications include underwater acoustic networks, UAV-based communication systems, and mobile robotic swarms—domains where conventional security methods are infeasible or insufficient. New approaches also aim to preserve location privacy while maintaining authentication strength, such as through obfuscated spatial signatures or spoof-resilient location sharing.

Chapter 3

Work Done

3.1 Uniform Linear Array

3.1.1 System Model

Consider a user equipment (UE), denoted Alice, equipped with a single antenna and located at coordinates $\mathbf{x}^A = [x_1^A, x_2^A]^\top$. The base station (BS), referred to as Bob, is positioned at the origin $[0, 0]^\top$ and features an analog array with $N > 1$ receive antennas and a single radio frequency (RF) chain. Alice transmits a sequence of $T > 1$ uplink pilot signals, represented as $\mathbf{s}^A = [s_1^A, \dots, s_T^A]^\top$, which satisfies the power constraint $\|\mathbf{s}^A\|^2 = 1$, where T is the number of pilot transmissions.

The signal received at Bob for each transmission $t = 1, 2, \dots, T$ is given by:

$$y_t = \sqrt{P} h^A \mathbf{w}_t^H \mathbf{a}(\theta^A) s_t^A + n_t, \quad (3.1)$$

where:

- P denotes the transmit power from Alice.
- h^A is the channel amplitude, defined as:

$$|h^A| = \frac{\lambda}{4\pi d^A}, \quad (3.2)$$

with $d^A = \|\mathbf{x}^A\|$ as the distance between Alice and Bob, and λ as the wavelength.

- $\mathbf{w}_t$ is the beamforming vector at time t , set to a directional beam, $\mathbf{w}_t = \mathbf{a}(\theta_t)$, where θ_t represents the probing direction within $[-90^\circ, 90^\circ]$.

- $\mathbf{a}(\theta^A)$ is the array steering vector, determined by the angle of arrival (AoA) $\theta^A = \arctan\left(\frac{x_1^A}{x_2^A}\right)$.
- n_t is complex additive white Gaussian noise (AWGN) with variance $\sigma^2/2$ per real dimension, where $\sigma^2 = N_0W$, with N_0 being the noise power spectral density and W the bandwidth.

It is assumed that Bob knows $\mathbf{s}^A$, enabling the estimation of the AoA $\hat{\theta}$ at the BS based on the received signals.

3.1.2 Impersonation Attacks

Authentication systems are vulnerable to attacks, particularly impersonation or spoofing attacks. We consider an attacker, Eve, equipped with a single-antenna user equipment (UE), who employs precoding to manipulate signals received by Bob, aiming to mislead Bob's classifier into identifying her signals as originating from Alice, as depicted in Fig. 1-(c).

General Attack Model

Eve is located at coordinates $\mathbf{x}^E = [x_1^E, x_2^E]^\top$, inducing an array steering vector $\mathbf{a}(\theta^E)$ at Bob, where the angle of arrival (AoA) is $\theta^E = \arctan\left(\frac{x_1^E}{x_2^E}\right)$. Eve manipulates the AoA-based authentication by transmitting a signal $\mathbf{s}^E = [s_1^E, \dots, s_T^E]^\top$, satisfying the power constraint $\|\mathbf{s}^E\|^2 = 1$. The signal received at Bob from Eve at time $t = 1, 2, \dots, T$ is modeled as:

$$y_t = \sqrt{P}h^E\mathbf{w}_t^H\mathbf{a}(\theta^E)s_t^E + n_t, \quad (3.3)$$

where:

- P is Eve's transmit power, assumed equal to Alice's for simplicity.
- h^E is the channel amplitude, given by $|h^E| = \frac{\lambda}{4\pi d^E}$, with $d^E = \|\mathbf{x}^E\|$ as Eve's distance from Bob.
- $\mathbf{w}_t$ is the beamforming vector, and $\mathbf{a}(\theta^E)$ is the array steering vector for Eve's AoA.
- n_t is complex additive white Gaussian noise (AWGN) with variance $\sigma^2/2$ per real dimension.

Eve manipulates s_t^E based on her knowledge of Alice's parameters, employing different strategies outlined below.

Random Attack

Eve knows the transmission timing and duration. She generates a signal $s_t^E = \frac{\exp(j\phi_t)}{\sqrt{T}}$, where $\phi_t \sim \mathcal{U}[0, 2\pi]$. Since $s_t^E \neq s_t^A$, this disrupts Bob's AoA estimation, but it is unlikely to enable Eve to impersonate Alice successfully.

Code-Based Attack

Eve knows the transmission timing, duration, beamforming vectors $\mathbf{w}_t$, pilot signals $\mathbf{s}^A$, and Alice's AoA θ^A , but not her own AoA θ^E . Eve manipulates her signal as:

$$s_t^E = \alpha^E \mathbf{w}_t^H \mathbf{a}(\theta^A) s_t^A, \quad (3.4)$$

where $\alpha^E = \left(\sum_{t=1}^T |\mathbf{w}_t^H \mathbf{a}(\theta^A) s_t^A|^2 \right)^{-1/2}$ ensures $\|\mathbf{s}^E\|^2 = 1$. The received signal at Bob is:

$$y_t = \sqrt{P} h^E \alpha^E \mathbf{w}_t^H \mathbf{a}(\theta^E) \mathbf{w}_t^H \mathbf{a}(\theta^A) s_t^A + n_t. \quad (3.5)$$

This causes Bob to perceive signals from both θ^E and θ^A , increasing the likelihood of Eve impersonating Alice.

Location-Based Attack

Eve has knowledge of $\mathbf{w}_t$, $\mathbf{s}^A$, θ^A , and her own AoA θ^E , derived from her location and Bob's. She manipulates her signal as:

$$s_t^E = \alpha^E \frac{\mathbf{w}_t^H \mathbf{a}(\theta^A) (\mathbf{w}_t^H \mathbf{a}(\theta^E))^*}{|\mathbf{w}_t^H \mathbf{a}(\theta^E)|^2} s_t^A, \quad (3.6)$$

where α^E is a normalization factor ensuring $\|\mathbf{s}^E\|^2 = 1$. The received signal at Bob becomes:

$$y_t = \sqrt{P} h^E \alpha^E \mathbf{w}_t^H \mathbf{a}(\theta^A) s_t^A + n_t. \quad (3.7)$$

This manipulation makes Bob perceive the signal as originating solely from θ^A , effectively masking θ^E . However, the attack's success depends on the power of α^E , which diminishes if $|\mathbf{w}_t^H \mathbf{a}(\theta^E)| \ll |\mathbf{w}_t^H \mathbf{a}(\theta^A)|$.

3.2 Uniform Planar Array

3.2.1 System Model

Consider Alice's single-antenna user equipment (UE) located at 3D coordinates $\mathbf{x}_A = [x_{A1}, x_{A2}, x_{A3}]^\top$. The base station (BS), corresponding to Bob, is located at the origin $\mathbf{0} = [0, 0, 0]^\top$, and is equipped with a Uniform Planar Array (UPA) consisting of $N = N_x \times N_y$ antennas, arranged in a two-dimensional grid along the x and y axes. The BS follows an analog beamforming architecture that consists of a single RF chain.

Alice transmits a sequence of $T > 1$ uplink pilot symbols $\mathbf{s}_A = [s_{A1}, \dots, s_{AT}]^\top$, normalized such that $\|\mathbf{s}_A\|^2 = 1$. The received signal at Bob for the t -th pilot transmission is modeled as:

$$y_t = \sqrt{P} h_A \mathbf{w}_t^H \mathbf{a}(\phi_A, \theta_A) s_{At} + n_t, \quad t = 1, 2, \dots, T \quad (3.8)$$

where:

- P denotes Alice's transmit power.
- $h_A = \frac{\lambda}{4\pi d_A}$ is the path loss-based channel amplitude, with λ as the carrier wavelength and $d_A = \|\mathbf{x}_A\|$ representing the Euclidean distance between Alice and the BS.
- $\mathbf{a}(\phi_A, \theta_A) \in \mathbb{C}^N$ is the UPA steering vector corresponding to Alice's azimuth $\phi_A \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ and elevation $\theta_A \in [0, \frac{\pi}{2}]$ angles of arrival (AoAs).
- $\mathbf{w}_t = \mathbf{a}(\phi_t, \theta_t)$ is the beamforming vector used to steer toward direction (ϕ_t, θ_t) at time t .
- $n_t \sim \mathcal{CN}(0, \sigma^2)$ denotes additive white Gaussian noise (AWGN).
- $\sigma^2 = N_0 W$, with N_0 as the noise power spectral density and W as the bandwidth.

The UPA array steering vector $\mathbf{a}(\phi, \theta)$ is defined as:

$$\mathbf{a}(\phi, \theta) = \exp(j\pi [r_x \sin \phi \cos \theta + r_y \sin \theta]), \quad (3.9)$$

where $r_x, r_y \in \mathbb{Z}$ denote the relative antenna indices along the x and y dimensions of the UPA, centered about the array origin.

To estimate the AoA pair $(\hat{\phi}, \hat{\theta})$, the BS evaluates a spatial likelihood function over a dense angular grid. By correlating the received signal sequence $\{y_t\}$ with beamformed

projections from a pre-defined 2D codebook of candidate angles, the direction pair with minimum residual error is selected as the AoA estimate.

3.2.2 Impersonation Attacks

Random Attack

In this strategy, Eve possesses limited knowledge—she knows the transmission time and duration but has no information about Alice’s pilot symbols, combining vectors, or AoA. Eve generates her signal vector

$$\mathbf{s}_E = [s_E^1, s_E^2, \dots, s_E^T]^\top$$

where each element is defined as:

$$s_E^t = \frac{1}{\sqrt{T}} \exp(j\phi_t), \quad \phi_t \sim \mathcal{U}[0, 2\pi]$$

This random phase modulation ensures that $\|\mathbf{s}_E\|^2 = 1$, satisfying the unit energy constraint. Due to the mismatch between Eve’s transmitted symbols and Alice’s pilot sequence, Bob’s AoA estimation is distorted, although impersonation remains statistically unlikely. The impact of the attack is primarily manifested as degradation in the reliability of AoA-based authentication, particularly in noisy conditions or when the antenna aperture is limited.

Location-Based Attack

This advanced strategy assumes that Eve has precise knowledge of:

- Her own location and resulting AoA (θ_E, ϕ_E) ,
- The BS’s combining vectors $\mathbf{w}_t$,
- Alice’s pilot symbols $\mathbf{s}_A$,
- And the target AoA (θ_A, ϕ_A) (azimuth and elevation angles for UPA).

To craft her signal, Eve exploits the structure of the UPA steering vector $\mathbf{a}(\theta, \phi)$, which accounts for both azimuth and elevation. The attack vector is computed as:

$$s_E^t = \alpha_E \cdot \mathbf{w}_t^H \mathbf{a}(\theta_A, \phi_A) \cdot \frac{(\mathbf{w}_t^H \mathbf{a}(\theta_E, \phi_E))^*}{|\mathbf{w}_t^H \mathbf{a}(\theta_E, \phi_E)|^2} \cdot s_A^t$$

where α_E is a normalization factor ensuring that $\|\mathbf{s}_E\|^2 = 1$. This sophisticated manipulation forces the effective signal received at Bob to appear as if it originated from Alice's direction:

$$y_t = \sqrt{P} \cdot h_E \cdot \alpha_E \cdot \mathbf{w}_t^H \mathbf{a}(\theta_A, \phi_A) \cdot s_A^t + n_t$$

This allows Eve to effectively impersonate Alice by projecting her signal onto the same spatial subspace as Alice's transmission. The UPA configuration enhances Eve's flexibility in beamforming, increasing the impersonation risk if the BS's angular resolution is insufficient.

3.3 Two Base Station

3.3.1 System Model

Let Eve be located at coordinates

$$\mathbf{x}_E = \begin{bmatrix} x_E^{(1)} \\ x_E^{(2)} \end{bmatrix}.$$

Eve induces an array steering vector $\mathbf{a}(\theta_E^{(i)})$ at BS $i \in \{1, 2\}$, determined by the angle-of-arrival (AoA)

$$\theta_E^{(i)} = \arctan \left(\frac{x_E^{(1)} - x_{B_i}^{(1)}}{x_E^{(2)} - x_{B_i}^{(2)}} \right).$$

Eve manipulates the AoA-based authentication process by transmitting

$$\mathbf{s}_E = [s_E^{(1)}, \dots, s_E^{(T)}]^\top, \quad \text{with } \|\mathbf{s}_E\|^2 = 1.$$

The received signal at BS i due to Eve's transmission at time t is given by:

$$y_t^{(i)} = \sqrt{P} h_E^{(i)} \mathbf{w}_t^{(i)H} \mathbf{a}(\theta_E^{(i)}) s_E^{(t)} + n_t^{(i)}, \quad (3.10)$$

where:

- $|h_E^{(i)}| = \frac{\lambda}{4\pi d_E^{(i)}}$, with $d_E^{(i)} = \|\mathbf{x}_E - \mathbf{x}_{B_i}\|$,
- $\mathbf{w}_t^{(i)} = \mathbf{a}(\theta_t^{(i)})$ is the beamforming vector,
- $n_t^{(i)} \sim \mathcal{CN}(0, \sigma^2)$ is AWGN.

3.3.2 Impersonation Attacks

Random Attack

Eve knows when to transmit and the duration of transmission, but has no knowledge of Alice's pilot or beamforming strategy. Eve transmits:

$$s_E^{(t)} = \frac{1}{\sqrt{T}} \exp(j\phi_t), \quad \phi_t \sim \mathcal{U}[0, 2\pi],$$

ensuring $\|\mathbf{s}_E\|^2 = 1$.

At BS i , the received signal becomes:

$$y_t^{(i)} = \sqrt{P} h_E^{(i)} \mathbf{w}_t^{(i)H} \mathbf{a}(\theta_E^{(i)}) \cdot \frac{1}{\sqrt{T}} e^{j\phi_t} + n_t^{(i)}.$$

Since $s_E^{(t)} \neq s_A^{(t)}$, AoA estimation at the BSs is compromised. However, due to the mismatch in pilot structure, impersonation of Alice remains unlikely.

Location-Based Attack

In this scenario, Eve knows:

- The beamforming vectors $\mathbf{w}_t^{(1)}$ and $\mathbf{w}_t^{(2)}$ used at BS1 and BS2,
- The pilot sequence $\mathbf{s}_A$ transmitted by Alice,
- The angles of arrival (AoAs) $\theta_A^{(1)}, \theta_E^{(1)}$ at BS1 and $\theta_A^{(2)}, \theta_E^{(2)}$ at BS2.

Eve crafts her signal to spoof both base stations simultaneously, manipulating each beam index t by sending:

$$s_E^{(t)} = \alpha_E \cdot \left[\frac{\mathbf{w}_t^{(1)H} \mathbf{a}(\theta_A^{(1)}) \left(\mathbf{w}_t^{(1)H} \mathbf{a}(\theta_E^{(1)}) \right)^*}{\left| \mathbf{w}_t^{(1)H} \mathbf{a}(\theta_E^{(1)}) \right|^2} \cdot \frac{\mathbf{w}_t^{(2)H} \mathbf{a}(\theta_A^{(2)}) \left(\mathbf{w}_t^{(2)H} \mathbf{a}(\theta_E^{(2)}) \right)^*}{\left| \mathbf{w}_t^{(2)H} \mathbf{a}(\theta_E^{(2)}) \right|^2} \right] \cdot s_A^{(t)},$$

where α_E is a normalization constant ensuring:

$$\alpha_E = \left(\sum_{t=1}^T \left| \frac{\mathbf{w}_t^{(1)H} \mathbf{a}(\theta_A^{(1)}) \left(\mathbf{w}_t^{(1)H} \mathbf{a}(\theta_E^{(1)}) \right)^*}{\left| \mathbf{w}_t^{(1)H} \mathbf{a}(\theta_E^{(1)}) \right|^2} \cdot \frac{\mathbf{w}_t^{(2)H} \mathbf{a}(\theta_A^{(2)}) \left(\mathbf{w}_t^{(2)H} \mathbf{a}(\theta_E^{(2)}) \right)^*}{\left| \mathbf{w}_t^{(2)H} \mathbf{a}(\theta_E^{(2)}) \right|^2} \right|^2 \right)^{-1/2}.$$

The received signal at base station $i \in \{1, 2\}$ is:

$$y_t^{(i)} = \sqrt{P_i} h_E^{(i)} \mathbf{w}_t^{(i)H} \mathbf{a}(\theta_E^{(i)}) \cdot s_E^{(t)} + n_t^{(i)}.$$

Substituting $s_E^{(t)}$, we obtain:

$$y_t^{(i)} = \sqrt{P_i} h_E^{(i)} \alpha_E \cdot \mathbf{w}_t^{(i)H} \mathbf{a}(\theta_A^{(i)}) \cdot s_A^{(t)} + n_t^{(i)}.$$

Hence, each base station perceives the signal as originating from $\theta_A^{(i)}$, effectively hiding Eve's location. However, the effective SNR is reduced by the normalization factor α_E , making Eve's attack detectable under cooperative detection schemes.

Chapter 4

Simulations and Results

4.1 Unifrom Linear Array

4.1.1 AoA Estimation

We adopt a maximum likelihood estimation (MLE) method to estimate the Angle of Arrival (AoA) from the received signal. Considering a sequence of T received pilot signals $y = [y_1, y_2, \dots, y_T]^\top$, the goal is to estimate the AoA θ_A in the presence of an unknown complex channel gain h_A .

Let the beamforming response be defined as:

$$z(\theta) = \begin{bmatrix} \mathbf{w}_1^H \mathbf{a}(\theta) \\ \mathbf{w}_2^H \mathbf{a}(\theta) \\ \vdots \\ \mathbf{w}_T^H \mathbf{a}(\theta) \end{bmatrix},$$

where $\mathbf{w}_t$ is the beamforming vector at time t , and $\mathbf{a}(\theta)$ is the array steering vector corresponding to angle θ .

The maximum likelihood estimate of the AoA is given by:

$$\hat{\theta} = \arg \min_{\theta} \left\| \mathbf{y} - \hat{h}(\theta) \mathbf{z}(\theta) \right\|^2,$$

where the estimate of h_A is:

$$\hat{h}(\theta) = \frac{\mathbf{z}^H(\theta) \mathbf{y}}{\|\mathbf{z}(\theta)\|^2}.$$

This estimator finds the AoA that minimizes the squared error between the observed signal and the best-fit spatial signal template parameterized by θ .

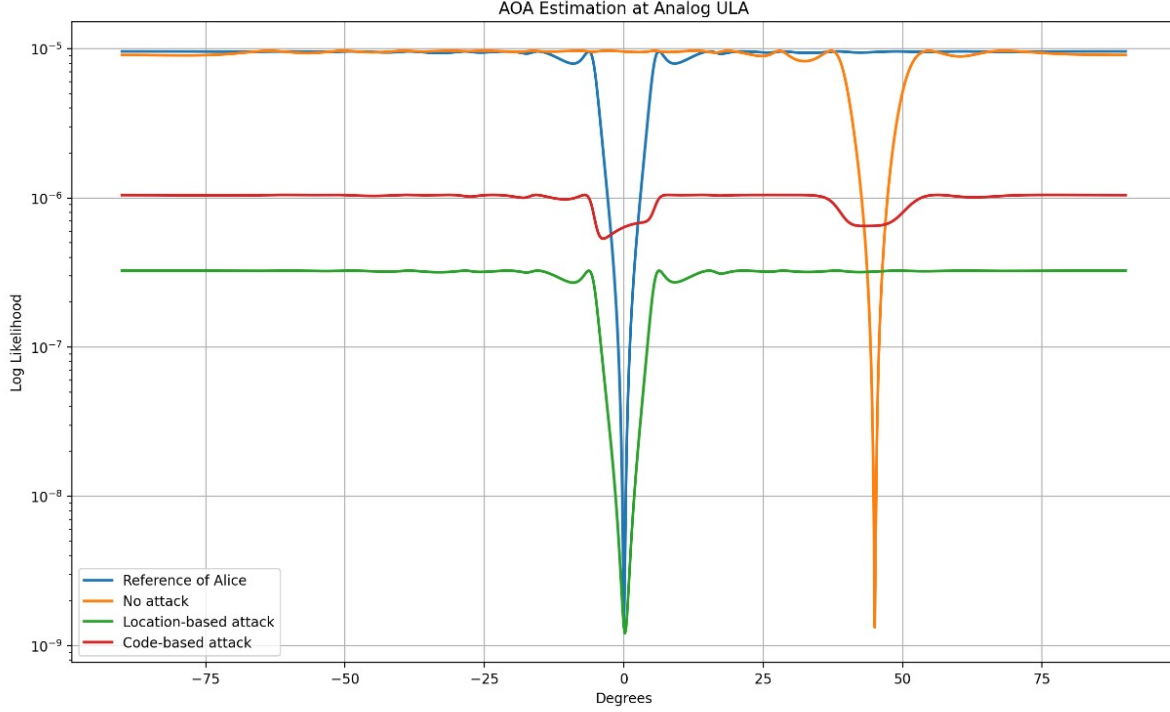
4.1.2 Simulation Parameters

The simulations are conducted with the following parameter settings:

- **Carrier frequency:** $f_c = 2.5$ GHz
- **Bandwidth:** $W = 20$ MHz
- **Transmit power:** $P = 10$ dBm
- **Noise power spectral density:** $N_0 = -174$ dBm/Hz
- **Base station antenna count:** $N = 16$
- **Number of pilot transmissions:** $T = 17$
- **Beamforming vectors:** $\mathbf{w}_t = \mathbf{a}(\theta_t)$, with θ_t uniformly spanning $[-90^\circ, 90^\circ]$
- **Alice's location:** distance $d_A = 10$ m, angle $\theta_A = 0^\circ$

The likelihood function derived from the MLE method is used to compute the angle estimates for both legitimate users and attackers under various attack scenarios. Monte Carlo simulations with 1000 trials are used to compute the root mean square error (RMSE) and classifier metrics.

4.1.3 Log Likelihood Plot



4.1.4 Impact of the Attacks on the Estimated AoA

To illustrate the effects of various impersonation attacks on AoA estimation, we analyze the negative log-likelihood cost function $\|\mathbf{y} - \hat{\mathbf{h}}(\theta)\mathbf{z}(\theta)\|^2$ over a range of angles θ . In the absence of any attack, this cost function exhibits a sharp minimum at the legitimate user's angle θ_A , indicating precise AoA estimation.

- **No Attack:** The cost function shows a distinct and sharp minimum at $\theta = \theta_A$, allowing reliable identification.
- **Random Attack:** The cost function is distorted and lacks a pronounced minimum around θ_A , preventing successful impersonation. However, repeated random attacks could potentially disrupt AoA estimation, leading to denial-of-service.
- **Code-based Attack:** A shallow secondary minimum appears near θ_A , while the main minimum remains at the attacker's actual angle θ_E . This indicates partial success in impersonation, but not a complete spoof.
- **Location-based Attack:** The original minimum at θ_E disappears, and a sharp new minimum appears at θ_A . This demonstrates that the location-based attack can fully

deceive the AoA estimator, making the received signal appear as if it originated from the legitimate direction.

This analysis confirms that the location-based attack is the most effective in terms of spoofing success, as it aligns the estimated AoA with the legitimate user's direction and suppresses any trace of the attacker's true location.

4.2 Uniform Planar Array

4.2.1 AoA Estimation

We adopt a *maximum likelihood (ML) estimation* framework to estimate the Angle of Arrival (AoA) of Alice, denoted as (ϕ_A, θ_A) , using signals received by a Uniform Planar Array (UPA). The presence of an unknown complex channel gain h_A leads to the following AoA estimation formulation:

$$\hat{\theta} = \arg \min_{\theta} \|\mathbf{y} - \hat{h}(\theta)\mathbf{z}(\theta)\|^2,$$

where $\mathbf{y} = [y_1, y_2, \dots, y_T]^T$ represents the received signal vector over T transmissions, and the beamforming projection vector $\mathbf{z}(\theta)$ is given by:

$$[z(\theta)]_t = \mathbf{w}_t^H \mathbf{a}(\theta),$$

with $\mathbf{w}_t$ as the transmit beamforming vector and $\mathbf{a}(\theta)$ the steering vector corresponding to a specific azimuth and elevation angle pair. The optimal estimate of the channel gain is given by:

$$\hat{h}(\theta) = \frac{\mathbf{z}^H(\theta)\mathbf{y}}{\|\mathbf{z}(\theta)\|^2}.$$

To analyze the impact of a *location-based spoofing attack*, we evaluate the *negative log-likelihood cost function* $\|\mathbf{y} - \hat{h}(\theta)\mathbf{z}(\theta)\|^2$ over a fine grid of azimuth and elevation angles.

- In the **benign scenario**, the cost function exhibits a **distinct sharp minimum at azimuth = 0° and elevation = 0°**, corresponding to Alice's true location.
- When **Eve is present without attack**, a **separate minimum appears at azimuth = 45° and elevation = 30°**, reflecting her actual spatial signature.

- However, when **Eve launches a location-based spoofing attack**, she adjusts her signal to emulate Alice’s spatial signature. As a result, the **minimum of the cost function shifts to azimuth = 0° and elevation = 0°**, while the genuine Eve’s location no longer appears as a distinct local minimum.

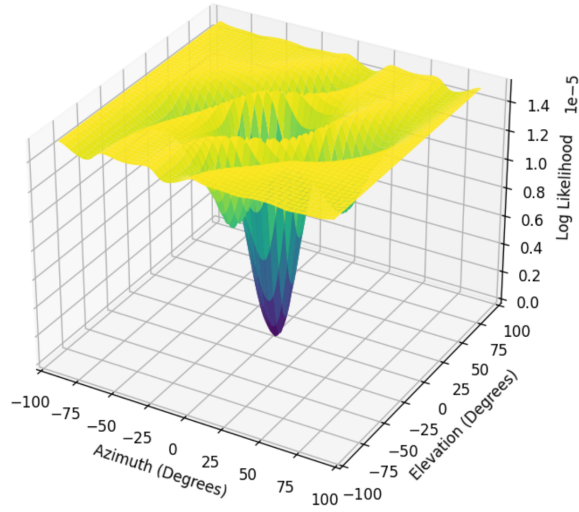
This behavior illustrates that the *location-based attack can successfully mislead the AoA estimator*, causing it to incorrectly attribute Eve’s transmission to Alice. This highlights a critical vulnerability in UPA-based physical-layer authentication systems and emphasizes the need for enhanced defense mechanisms incorporating multi-dimensional spatial features or cross-layer information.

4.2.2 Simulation Parameters

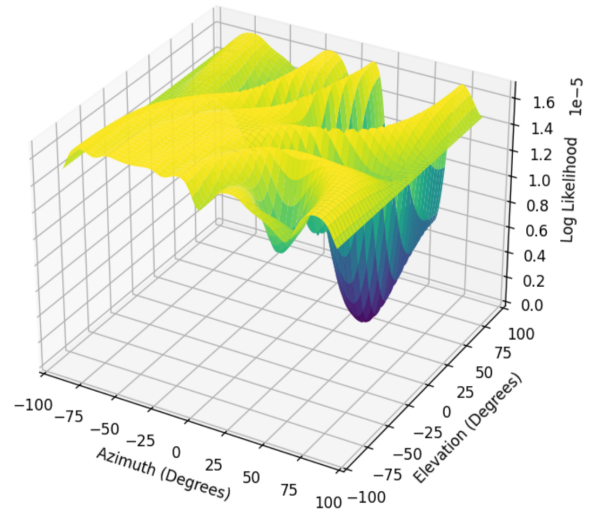
The simulations are conducted with the following parameter settings:

- **Carrier frequency:** $f_c = 2.5$ GHz
- **Bandwidth:** $W = 20$ MHz
- **Transmit power:** $P = 10$ dBm
- **Noise power spectral density:** $N_0 = -174$ dBm/Hz
- **Base station antenna count:** $N = 16$, arranged in a 4x4 grid
- **Number of pilot transmissions:** $T = 17$
- **Beamforming vectors:** $\mathbf{w}_t = \mathbf{a}(\theta_t)$, with θ_t uniformly spanning $[-90^\circ, 90^\circ]$
- **Alice’s location:** distance $d_A = 10$ m, angle $\theta_A = 0^\circ$
- **Eve’s location:** distance $d_A = 10$ m, angle $\theta_A = 30^\circ$
- The simulation uses a fine angular grid of **1501** points for accurate resolution of spatial signatures.

4.2.3 Log Likelihood Plots



(a) Reference of Alice



(b) No Attack

Figure 4.1: AOA Estimation Comparisons

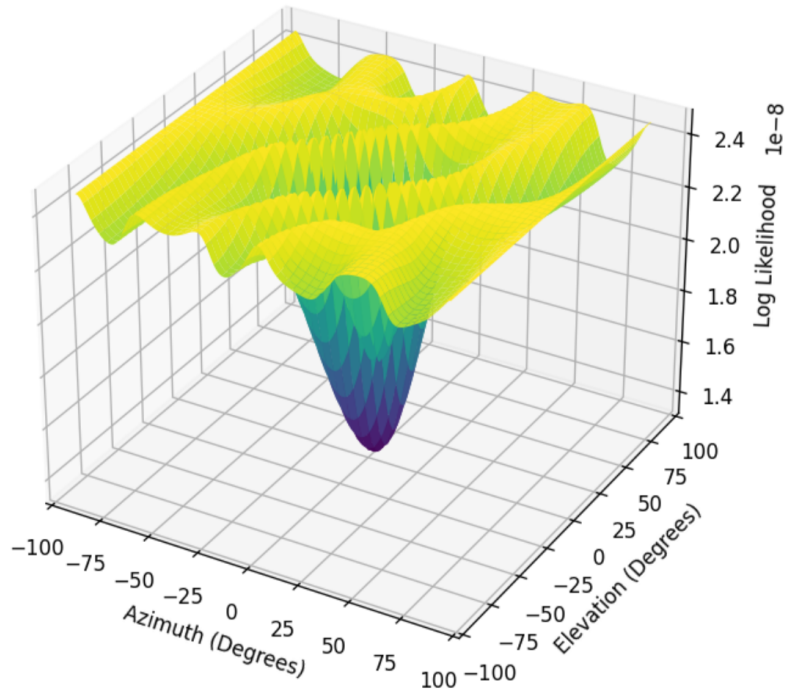


Figure 4.2: Location Based Attack

4.3 Two Base Station

4.3.1 AoA estimation

The angle of arrival (AoA) estimation in our two-base station setup is based on the *maximum likelihood (ML)* estimator, and extended to operate independently at each base station (BS1 and BS2). For a given angle θ , we define the beamforming response as $\mathbf{z}(\theta) = [w_1^H a(\theta), \dots, w_T^H a(\theta)]^\top$, where w_t are the directional beamformers and $a(\theta)$ is the array steering vector. The received signal over T beam directions is given by $\mathbf{y} = [y_1, y_2, \dots, y_T]^\top$. The complex channel gain h is unknown and estimated for each angle as

$$\hat{h}(\theta) = \frac{\mathbf{z}^H(\theta)\mathbf{y}}{\|\mathbf{z}(\theta)\|^2}.$$

The AoA estimate is then obtained by minimizing the squared error between the received signal and its projection:

$$\hat{\theta} = \arg \min_{\theta} \left\| \mathbf{y} - \hat{h}(\theta)\mathbf{z}(\theta) \right\|^2.$$

This estimation is performed independently at both BS1 and BS2, yielding AoA estimates $\hat{\theta}_1$ and $\hat{\theta}_2$. For a legitimate transmitter (Alice), these estimates are consistent with her known location, whereas an attacker (Eve) attempting to mimic Alice's signal will generally produce mismatched AoAs at the two BSs due to spatial separation and beamforming limitations. Thus, by comparing $\hat{\theta}_1$ and $\hat{\theta}_2$, we can identify impersonation attempts. This dual-BS likelihood-based estimation approach significantly enhances the reliability and security of AoA-based physical layer authentication.

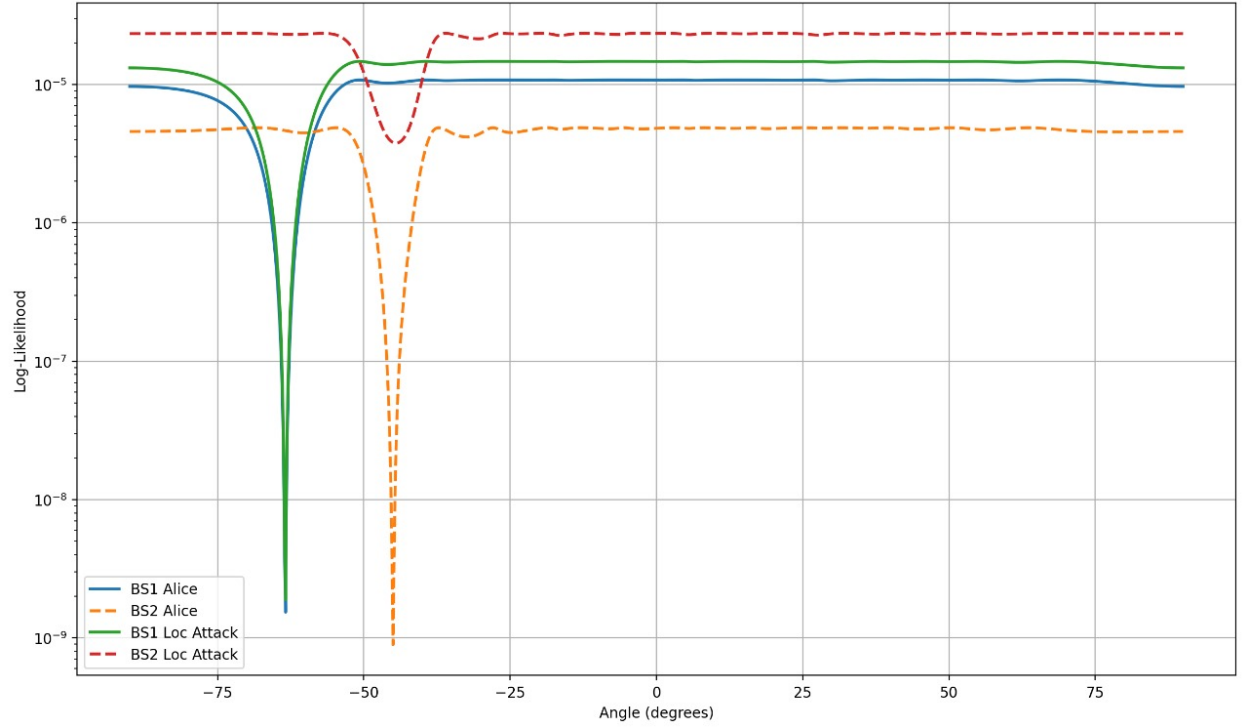
4.3.2 Simulation Parameters

The simulations are conducted with the following parameter settings:

- **Operation frequency f_c :** 2.5 GHz
- **Bandwidth (BW) f_c :** 20 MHz
- **Transmit Power $P_d B$:** 10 dBm
- **Noise Power Spectral Density $N_0 B mHz$:** -174 dBm/Hz
- **Number of antennas N :** 16 per base station

- Number of Transmissions / Beamforming Directions T : 17
- Beamforming Vectors $\mathbf{w}_t : \mathbf{a}(\theta_t)$ (Directional beams using array response vector)
- Probing Directions (θ_t) : Uniformly spaced from -90° to 90°
- Number of base stations: 2
- BS1 position: (5, 10)
- BS2 position: (0, 10)
- Alice (legitimate user) position: (10, 0)
- Eve (spoofer) position: variable

4.3.3 Log Likelihood Plot



Chapter 5

Challenges and Future Scopes

5.1 Challenges

5.1.1 Two Base Station Case

- **Interference and Noise Sensitivity:** AoA estimation accuracy heavily depends on signal-to-noise ratio (SNR). Under low SNR or severe multipath conditions, even a dual-BS setup may struggle to correctly identify impersonation.
- **Synchronization and Calibration:** Operating two base stations requires tight synchronization in time and frequency domains, along with precise antenna calibration to avoid angular misestimation.
- **Computational Overhead:** Performing likelihood-based AoA estimation over large angular grids across two BSs significantly increases computational complexity and latency.

5.1.2 3D Case

- **Increased Dimensional Complexity:** Estimating both azimuth and elevation angles expands the search space, requiring denser 2D angular grids and higher computation.
- **Antenna Array Design:** Uniform linear arrays (ULAs) only support azimuth estimation. 3D AoA estimation demands more complex arrays like uniform planar arrays (UPAs) or circular arrays that require advanced calibration.

- **Angle Ambiguity and Side Lobes:** In 3D space, multiple directions may produce similar responses due to side lobes or symmetrical patterns, leading to estimation ambiguities.

5.2 Conclusion

5.2.1 Two Base Station Case

The use of two base stations for AoA-based physical layer authentication offers significant improvements in detecting impersonation attacks compared to a single-BS setup. By independently estimating the angle of arrival at each BS and comparing the results, the system can detect inconsistencies introduced by adversaries. This cooperative detection leverages spatial diversity and geometric constraints, making it more resilient against attackers who may fool one BS but not both. Simulation results demonstrate that this dual-BS framework improves estimation accuracy and classification confidence, particularly in the presence of location-based spoofing.

5.2.2 3D Case

Extending AoA-based authentication into the 3D domain enhances its capacity to differentiate between legitimate and malicious users, especially in complex and non-line-of-sight (NLoS) scenarios. By estimating both azimuth and elevation angles from multiple base stations, a complete 3D spatial signature of the transmitter can be formed. This significantly improves the detection of sophisticated spoofing attempts. Simulation results indicate that 3D AoA reduces impersonation success rates and increases the resolution for distinguishing between closely spaced users. Although more computationally intensive, the 3D approach provides a higher security guarantee in real-world deployments.

5.3 Future Scope

5.3.1 Two Base Station Case

- **Multi-BS Fusion Techniques:** Expanding the system to incorporate more than two base stations enables triangulation and sensor fusion methods like Kalman filtering to estimate transmitter positions in addition to AoA.

- **Multipath and NLoS Environments:** Future work can explore how the system performs in urban or indoor environments, where multipath effects and non-line-of-sight conditions dominate.

5.3.2 3D Case

- **Full 3D Localization and Tracking:** Integrating AoA with time-of-arrival (ToA) or time-difference-of-arrival (TDoA) measurements can enable real-time 3D positioning of both users and adversaries.
- **Drone and UAV Applications:** Applying 3D AoA-based authentication to aerial networks, such as UAV swarms, can enhance security where elevation angles play a crucial role in verifying identities.

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